An Exact Affine Perpetuity and Palm Occupation Certificate for Linear-Rate AIMD

HCS Research Program

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Abstract

We study the hybrid process $\dot{X} = a > 0$ with jump intensity ρX and multiplicative decrease $X \mapsto \beta X$, $0 < \beta < 1$. For the pre-jump value Y_n, completing the integrated hazard gives the exact square-affine chain $Y_{n+1}^2 = \beta^2 Y_n^2 + 2aE_{n+1}/\rho$. This yields a unique stationary affine perpetuity, a source-local Laplace q-product, all squared moments, the generator moment recursion, and a stationary Markov-renewal/Palm occupation formula. The receipt covers 27 rational parameter tuples and exact reward skeletons, with independent checking, symbolic identities, replay, and hostile mutations. For positive β the jump sequence is Markov rather than iid regenerative; $\beta = 0$ is the separate reset face. No arithmetic target determinant is claimed, so the strict Route-A verdict is `ROUTE_A_REJECTED`.

1 Process and the pre-jump chain

Between jumps let $X_t = \beta Y_n + at$, where Y_n is the value immediately before the preceding jump. The next waiting time T_{n+1} is determined by the integrated hazard

$$\rho\left(\beta Y_n T_{n+1} + \frac{aT_{n+1}^2}{2}\right) = E_{n+1}, \qquad E_{n+1} \sim \mathrm{Exp}(1), \tag{1}$$

and $Y_{n+1} = \beta Y_n + aT_{n+1}$.

Theorem 1 (Square-affine recurrence)**.** *For every $a, \rho > 0$ and $0 < \beta < 1$,*

$$Y_{n+1}^2 = \beta^2 Y_n^2 + \frac{2a}{\rho} E_{n+1}. \tag{2}$$

Proof. Expand $(\beta Y_n + aT)^2 - \beta^2 Y_n^2 = 2a(\beta Y_n T + aT^2/2)$ and use (1). The factor a cannot be dropped. ☐

Theorem 2 (Perpetuity and uniqueness)**.** *With $Z_n = Y_n^2$, synchronous coupling under common exponentials satisfies $|Z_n - Z_n'| = \beta^{2n}|Z_0 - Z_0'|$. Consequently*

$$Z_\infty = \frac{2a}{\rho}\sum_{j\ge 0} \beta^{2j} E_{-j}$$

converges almost surely and is the unique stationary jump-chain law; every finite initial state converges to it.

Proof. The geometric coefficients have finite sum $\mathbb{E}Z_\infty = (2a/\rho)/(1-\beta^2) < \infty$, so the nonnegative perpetuity series is finite almost surely. The coupling identity contracts every initial pair, so two stationary versions coincide and the finite-start law converges weakly (indeed in Wasserstein metrics with a finite first moment). ☐

2 Laplace product and moments

Independence of the exponential innovations gives the source-local transform

$$\psi(s) = \mathbb{E}[e^{-sZ_\infty}] = \prod_{j\ge 0}\left(1 + \frac{2a}{\rho}\beta^{2j}s\right)^{-1}. \tag{3}$$

The release stores a 12-factor rational prefix and three rational spot values for every frozen parameter tuple. Equation (3) is probability bookkeeping, not an arithmetic Euler product or a target determinant.

For the continuous-time stationary law let $\varphi(s) = \mathbb{E}[e^{-sX}]$. Applying the generator (6) to e^{-sx} gives the source-local Laplace-generator identity

$$\varphi'(s) - \varphi'(\beta s) = \frac{a}{\rho} s\,\varphi(s). \tag{4}$$

It is not a target continuation or functional equation.

Writing $m_k = \mathbb{E}[Z^k]$, expansion of (2) yields

$$(1-\beta^{2k})m_k = \sum_{j=0}^{k-1} \binom{k}{j} \beta^{2j} \left(\frac{2a}{\rho}\right)^{k-j} (k-j)! m_j, \qquad m_0 = 1. \tag{5}$$

All entries in the finite receipt are exact rational numbers.

3 Stationary occupation formula

The continuous-time generator is

$$(Lf)(x) = af'(x) + \rho x \left[f(\beta x) - f(x)\right]. \tag{6}$$

For stationary moments $M_m = \mathbb{E}[X^m]$, applying (6) to x^m gives

$$\rho(1-\beta^m)M_{m+1} = amM_{m-1}, \qquad m \geq 1. \tag{7}$$

Thus $M_0 = 1$, $M_1 = \mu$, and all higher moments are exact affine expressions in the single Palm-determined mean μ.

Let π denote the stationary pre-jump law. A cycle from βY_n to Y_{n+1} has duration $T = (Y_{n+1} - \beta Y_n)/a$ and monomial reward

$$\int_0^T (\beta Y_n + at)^m \, dt = \frac{Y_{n+1}^{m+1} - (\beta Y_n)^{m+1}}{a(m+1)}. \tag{8}$$

The stationary Markov-renewal/Palm ratio is therefore

$$M_m = \frac{1-\beta^{m+1}}{(m+1)(1-\beta)} \frac{\mathbb{E}_\pi[Y^{m+1}]}{\mathbb{E}_\pi[Y]}, \qquad \mu = \frac{a}{\rho(1-\beta)\mathbb{E}_\pi[Y]}. \tag{9}$$

For $\beta > 0$, π is a stationary Markov jump-cycle law: (8) is not an iid regeneration assertion.

References

[1] V. Dumas, F. Guillemin, and P. Robert, "A Markovian analysis of additive-increase multiplicative-decrease algorithms," *Advances in Applied Probability* 34 (2002), DOI: 10.1239/aap/1019160951.

[2] F. Guillemin, P. Robert, and B. Zwart, "AIMD algorithms and exponential functionals," *Annals of Applied Probability* 14 (2004), DOI: 10.1214/aoap/1075828048.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no arithmetic, target-zero, Euler-factor, root-data, automorphy, or Hilbert–Pólya claim is made. **Data and code.** Exact receipts, independent checker, replay, and mutation audit are included. **AI-use disclosure.** Generative tools assisted drafting and code generation; formulas and metadata are checked by the deterministic artifact chain.